

Challenge 1

Build a ETL

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Constitution

Mision

A company need to applies a 10% discount to the first 500 online orders placed through the platform. Only the first order per customer per day counts toward the threshold.

Tech Stack

- Data extraction via kaggle hub**
- Jupyter Notebook built with Python (pandas, NumPy, SQLAlchemy)**
- PostgreSQL database hosted on Neon serverless, connected via psycopg2**

Roadmap

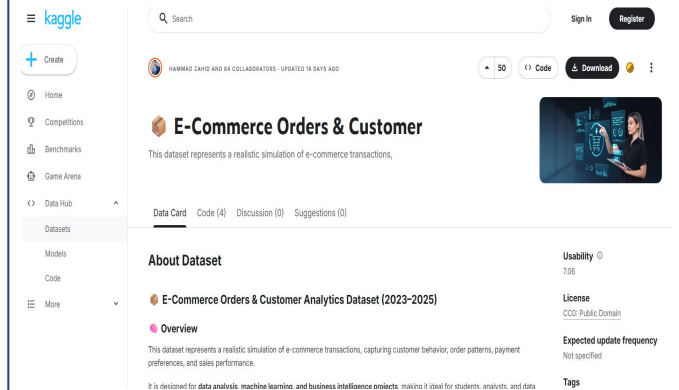
- Extract the dataset from the Kaggle platform**
- Build a Jupyter notebook ETL pipeline (Extract → Transform → Load)**
- Apply discount business logic and load data into a Neon PostgreSQL database using a medallion architecture (Bronze / Silver / Gold)**

Feature Phase

Phase 1

Task 1

Download the e-commerce dataset from Kaggle using kaggle hub and cache it locally



The screenshot shows the Kaggle interface for the 'E-Commerce Orders & Customer' dataset. The left sidebar contains navigation links: Home, Competitions, Benchmarks, Game Arena, Data Hub, and Datasets. The main content area displays the dataset title, a description, and a 'Download' button. Below the title, there are tabs for 'Data Card', 'Code (4)', 'Discussion (0)', and 'Suggestions (0)'. The 'About Dataset' section provides more details about the dataset's origin and purpose.

E-Commerce Orders & Customer

This dataset represents a realistic simulation of e-commerce transactions, capturing customer behavior, order patterns, payment preferences, and sales performance.

About Dataset

E-Commerce Orders & Customer Analytics Dataset (2023-2025)

Overview

This dataset represents a realistic simulation of e-commerce transactions, capturing customer behavior, order patterns, payment preferences, and sales performance.

Usability

7.06

License

CC0: Public Domain

Expected update frequency

Not specified

Tags

Task 2

Rename columns, cast types, deduplicate by (date, customer), and apply the 10% discount logic for the first 500 online orders.

Feature Phase

Phase 1

Task 3

Write data through a medallion architecture:

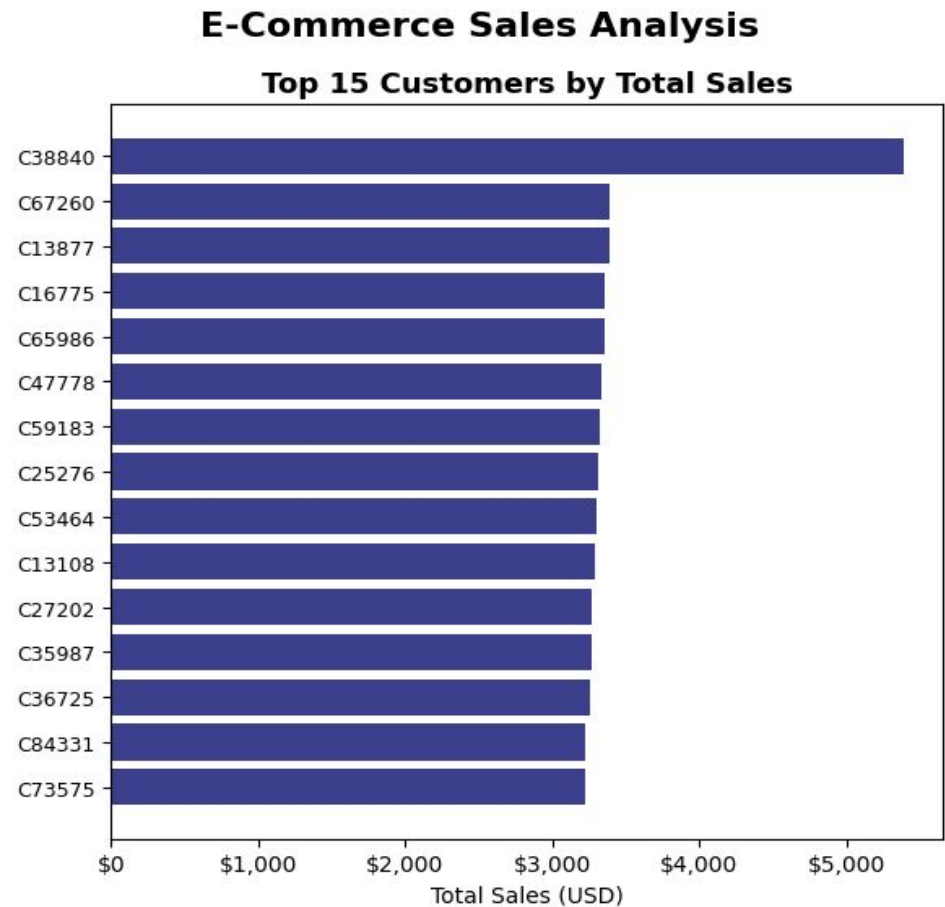
- **Bronze:** raw deduplicated data (raw_ventas)
- **Silver:** enriched with discount columns (silver_ventas)
- **Gold:** star schema (dim_cliente, dim_producto, fact_ventas)

Task 4

Query the star schema on Neon for business metrics: sales by customer, by product, and by day.

Total sales by customer

Ranks customers by total amount spent (final_total) across all their orders.

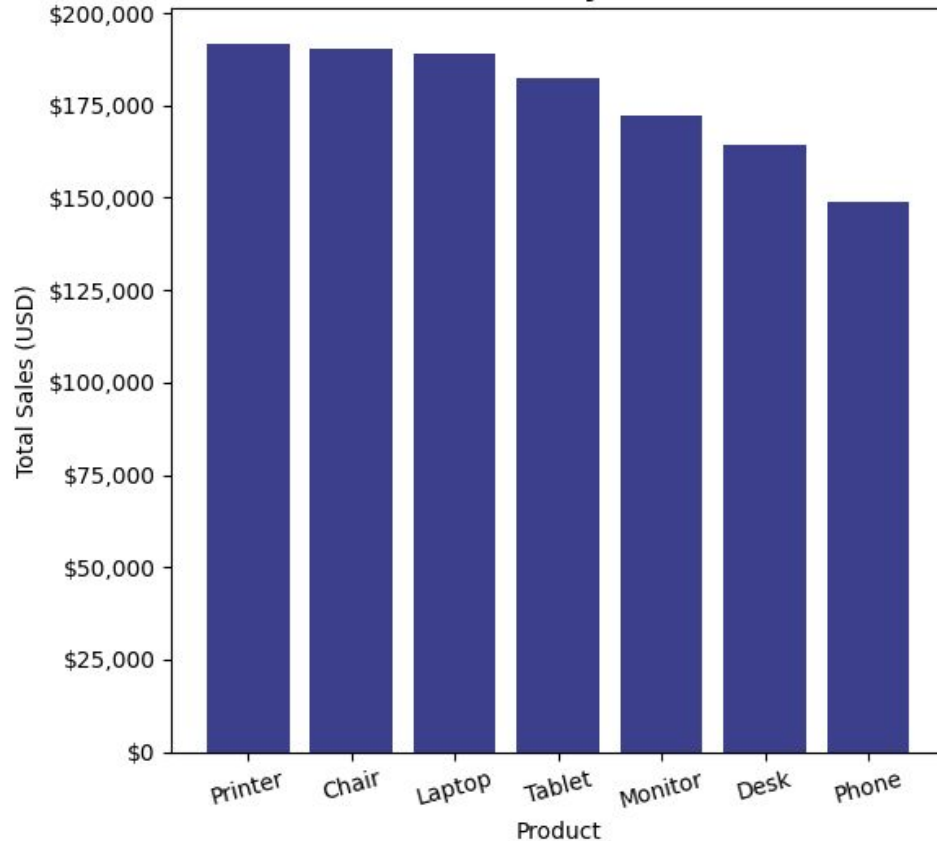


Total sales by product

Ranks products by total revenue generated

E-Commerce Sales Analysis

Total Sales by Product

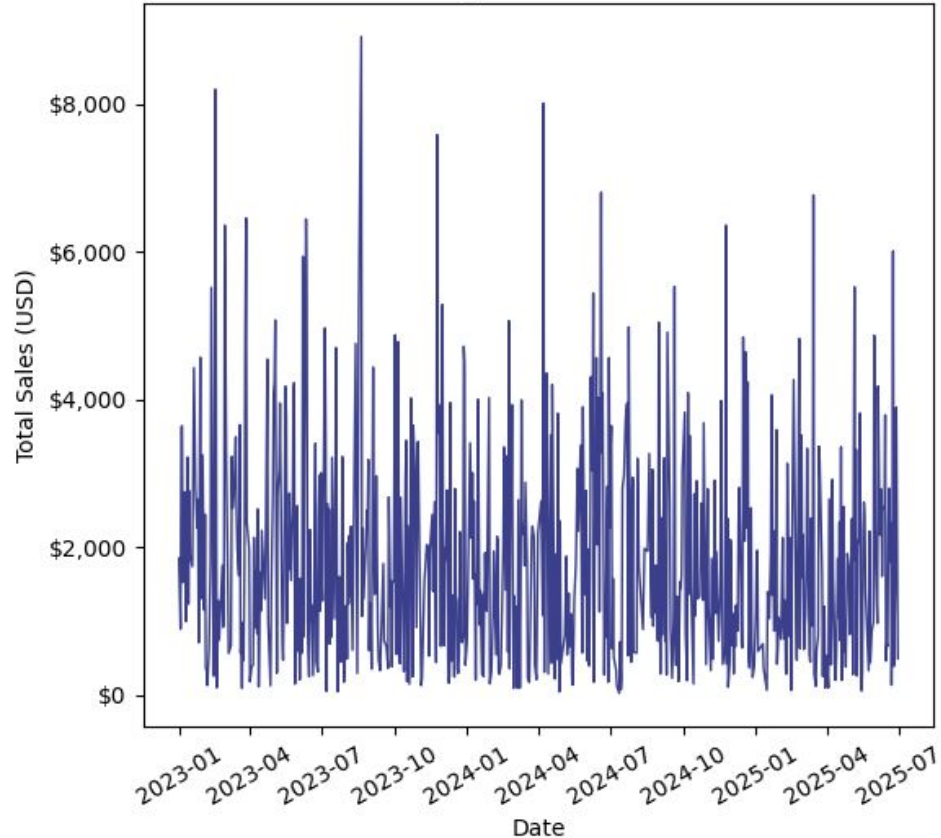


Sales by day

Daily revenue totals ordered chronologically – useful for trend and seasonality analysis.

E-Commerce Sales Analysis

Daily Sales Trend





Thanks